
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Notwithstanding their impressive empirical performance, generative image models trained on large-scale corpora recurrently fail to render seemingly elementary concepts—for instance, human hands or objects appearing in groups of four—whose presence in the training distribution one would reasonably presuppose. Hitherto, such failure modes have been chronicled predominantly through anecdote, thereby leaving unresolved the question of whether they constitute idiosyncratic aberrations or, instead, manifestations of more structural limitations intrinsic to these models. To adjudicate this question, we introduce a systematic methodology for the identification and characterization of “conceptual blindspots”—concepts that, though present in the training data, are either absent from or misrepresented within a model’s generations. Our approach harnesses sparse autoencoders (SAEs) to distil interpretable concept embeddings, thereby enabling a quantitative comparison of concept prevalence between real and generated imagery. We train an archetypal SAE (RA-SAE) on DINOv2 features endowed with 32,000 concepts—the largest such SAE constructed to date—which affords a fine-grained interrogation of conceptual disparities. Deployed across four widely adopted generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach surfaces specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) alongside exaggerated blindspots (e.g., wood background texture and palm trees). At the granularity of individual datapoints, we further isolate memorization artifacts—instances in which models reproduce highly specific visual templates encountered during training. Taken together, we advance a theoretically grounded framework for the systematic identification of conceptual blindspots in generative models, predicated on an assessment of their conceptual fidelity relative to the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, thereby underwriting their deployment in applications situated both within the vision domain [1–9] and considerably beyond it [9–15]. This success notwithstanding, a body of qualitative [16–18] and quantitative investigations [19, 20] has demonstrated that these models can, on occasion, falter in synthesizing images depicting comparatively rudimentary concepts—for instance, human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Indeed, when prompted to generate images encompassing such concepts, models are inclined to yield outputs bearing related structures while nonetheless departing from the precise ground-truth concept—rendering, say, hands endowed with six fingers. These failure



Figure 1: $\delta(k)$ quantifies a model’s propensity to over- or under-generate a concept c_k relative to its frequency in natural data. We deem concepts for which $\delta(k) < 0.1$ to constitute **suppressed conceptual blindspots** and those for which $\delta(k) > 0.9$ to constitute **exaggerated conceptual blindspots**. The depicted images, generated by four widely adopted generative image models, exemplify images exhibiting conceptual blindspots, together with instances of aligned concepts.

37 modes, which we designate “conceptual blindspots”¹ prove counterintuitive, inasmuch as one might
 38 reasonably anticipate that these models were exposed to a sufficiency of demonstrations faithfully
 39 articulating such concepts. This state of affairs raises the question of whether such failures betray
 40 intriguing idiosyncrasies confined to a handful of specific concepts, or whether they instead evidence
 41 a more systematic phenomenon whereby, across a broad spectrum of concepts, models are either
 42 disinclined or, conversely, excessively prone to produce images containing them.

43 Adjudicating this question remains infeasible under existing approaches for evaluating generative
 44 image models [28, 29]. Specifically, prevailing approaches typically rest upon coarse-grained
 45 measures devised to appraise image realism—for example, FID [30]—and consequently fail to
 46 apprehend distributional shortcomings. Methods such as CLIPScore, which quantify generation
 47 diversity or distribution-coverage statistics [31–33], furnish only partial purchase on our question,
 48 since they operate at neither the granularity of fine-grained features nor that of concepts [34, 35],
 49 thereby rendering the identification of conceptual blindspots difficult. Finally, qualitative modes of
 50 evaluation—such as participant surveys [5, 17, 36–38] or open-ended exploration [39]—can indeed
 51 surface deficiencies in a model’s capacity to capture particular concepts, yet they afford no scalable
 52 methodology amenable to ready replication across model families and to their systematic comparison.

53 **This work.** Motivated by the foregoing considerations, we contend that the identification and analysis
 54 of conceptual blindspots in a generative image model necessitates a methodology capable of eliciting,
 55 in an automated and unsupervised fashion, those concepts within the data distribution whose odds of
 56 generation under the true data-generating process are at variance with those under the trained model.
 57 Our contributions in this work are three-fold:

- 58 • **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a systematic
 59 framework for the identification and quantification of conceptual blindspots in generative image
 60 models relative to natural images (Section 3). This formalization transcends anecdotal or human-
 61 defined concept evaluations, thereby furnishing a principled apparatus for apprehending the
 62 models’ limitations.
- 63 • **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse
 64 Autoencoders.** We develop a structured methodology, grounded in sparse autoencoders (SAEs),
 65 for extracting and comparing concept distributions across natural and synthesized images (Sec-
 66 tion 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features
 67 endowed with 32,000 concepts—the largest such RA-SAE constructed to date.
- 68 • **Exploratory Tool and Analysis.** Our exploratory web tool underwrites both distribution- and
 69 datapoint-level analysis of blindspots across models (Sections 5.1–5.4). We apply our method
 70 to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). Therein we isolate spe-
 71 cific instances of both suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and
 72 whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background
 73 texture and palm trees), as depicted in Fig. 1

74 The model, code, and web tool are made available at a software repository.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.

75 2 Related Work

76 2.1 Explainability in Vision

77 Early work in explainable AI, computer vision included, concentrated upon methods for attributing
78 predictions to influential input regions [40–42]. These methods, however, conveyed only limited
79 semantic information regarding learned representations and not infrequently produced erroneous
80 explanations [43–45]. To redress these deficiencies, concept-based interpretability [46] emerged with
81 the aim of identifying semantically meaningful directions within neural networks, thereby revealing
82 not merely where such networks attend but which concepts and structures they marshal [47–49].

83 Recent work establishes that the prevailing concept-based interpretability methods—ACE [44],
84 CRAFT [48], and Sparse Autoencoders (SAEs) [50–51]—in essence confront the selfsame dictionary
85 learning task, differing only in the constraints they impose [52]. Among these approaches, sparse
86 autoencoders (SAEs) have come to the fore as particularly scalable for concept-based interpretability.
87 Although recent studies have laid bare certain limitations of the original SAEs—among them exces-
88 sively specific features [53], difficulties with compositionality [54], and circumscribed intervention
89 effects [55]—refined variants have since been proposed, notably the archetypal SAE (RA-SAE) [56].

90 Beyond SAEs, a further repertoire of interpretability methods—encompassing prompt-based prob-
91 ing [57], visualizations of attention maps or activations [58–59], and dataset-level statistics [32–33]
92 (e.g., diversity or distribution-coverage metrics)—affords only partial purchase on these questions.
93 Crucially, such methods want granularity, fixating upon whole images or prompts rather than upon
94 fine-grained features and concepts [34–35]. Moreover, they are beholden to subjective interpretation
95 and fail to discriminate among the various modes of failure [60]. The extant methods and metrics are,
96 in consequence, inadequate to the systematic identification of feature- and concept-level infirmities in
97 generative image models [28].

98 2.2 Generative Image Models

99 Diffusion-based methods have attained a position of dominance across the diverse modalities of
100 generative vision modeling, encompassing image [4–61–64], video [65–72], and 3D [7–73–75]
101 synthesis. Within the domain of image generation, this lineage may be traced to denoising diffusion
102 probabilistic models (DDPMs) [76], subsequently generalized to non-Markov diffusion processes
103 through denoising diffusion implicit models (DDIMs) [77].

104 The Stable Diffusion (SD) [9] model family rendered DDPMs eminently accessible to the research
105 and open-source communities alike. The original SD was succeeded by a series of subsequent
106 iterations, among them SD 2 [78], SD 3 [79], and SD XL [80]. A profusion of modifications and
107 extensions of the SD architecture has since arisen, admitting supplementary constraints upon the
108 diffusion process (e.g., style [81–82], pose [83], and identity [84–85]) as well as alternative input
109 modalities, such as image-to-image generation. A variety of Latent Diffusion Models erected atop
110 SD—including Kandinsky [86], PixArt [87], DeepFloyd [4–88], and FLUX [89]—have likewise
111 emerged.

112 2.3 Datasets for Generative Image Models

113 The recent success of generative vision models is attributable in large measure to the concomitant
114 abundance of computational resources and large-scale internet datasets [90–92]. LAION-5B [93],
115 in particular, has assumed a pivotal role in the training of open-source text-to-image models, SD
116 and its derivatives among them. Scraped from Common Crawl [94], this dataset comprises upward
117 of 5B image-caption pairs, of which 2.3B are English-language. Other prominent datasets include
118 COYO-700M [95] and Conceptual Captions [96], comprising 700M and 3M image-caption pairs,
119 respectively.

120 As LAION-5B gained currency, apprehensions concerning its biases mounted [97–101]. Notwith-
121 standing attempts at filtering, harmful content persisted [97–99]—including NSFW material [100]
122 and several hundred instances of CSAM [101]—precipitating the dataset’s temporary withdrawal
123 from official channels. The dataset is further afflicted by low-quality images [102] and by internet-
124 style captions (e.g., product descriptions) that stand in misalignment with the manner in which users
125 prompt trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations of input data, wherein each sample admits reconstruction as a linear combination of a small number of dictionary atoms [104–106]. Erected upon the theory of compressed sensing [107–108], the field progressed from early vector quantization methods [109] toward more sophisticated formulations, including Non-negative Matrix Factorization [110–111], Sparse PCA [112], and K-SVD [113]. More recent advances comprise online methods [114], structured sparsity [115], and theoretical guarantees [116–117], in tandem with an increasingly intimate rapprochement with deep learning [118, 119].

Advances in sparse coding have moreover been enlisted to interrogate the emergence of high-level concepts within diffusion models [120]. Antecedent to diffusion models, concept-grounded interpretability was brought to bear upon earlier generative architectures by way of concept-bottleneck models, which exact human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both approaches, however, are predicated upon human-defined concepts and thus inevitably overlook broader trends that the user does not explicitly register.

Aggregate metrics (e.g., precision, recall, density, and coverage [35, 123]) together with latent density scores, which forecast sample quality on the basis of the model’s latent space [124], have arisen to appraise the capabilities of generative image models. Although these effectively expose distributional lacunae, they yield scant insight into the specific concepts that are under- or over-represented.

Cumulatively, these limitations motivate a scalable, unsupervised framework capable of systematically identifying and quantifying concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We commence by formalizing the notion of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. This formulation empowers us to derive principled, quantitative measures that characterize which concepts a generative model under- or over-represents relative to its data distribution. The process is illustrated in Fig. 2. Although we predicate this pursuit upon standard assumptions [126–130], we find empirically that meaningful phenomenology is elicited even where these assumptions are contravened.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational parsimony, we employ $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This conflation is warranted by the invertibility of G , which, via the change-of-variables formula, induces a valid distribution over $\mathcal{X}$.

In essence, the individual dimensions of the latent space instantiate the **Concepts** that underlie the data distribution $P_{\mathcal{X}}$, itself defined over some observation space of images $\mathcal{X}$. By way of illustration, distinct latents may correspond to concepts such as color, shape, size, location, and the like [131–132]. We further permit the data-generating process to associate a text-description $t \in \mathcal{T}$ with any image sampled from the data distribution, though we refrain from modeling it explicitly. These text descriptions may thereafter be enlisted to train a text-conditioned **Generative image model** g_{θ} , parameterized by θ , upon a corpus of image-text pairs, so as to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, I)$ together with a text-description of the scene t to images x that render the latter.

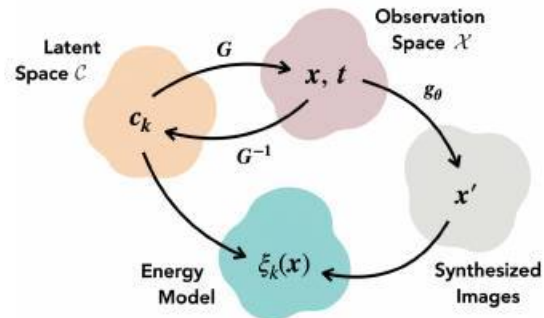


Figure 2: Intuitive mapping of our framework. Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP, whereupon S_{θ} synthesizes images x' conditioned on t . The energy model $\xi_k(x)$ then extracts concept representations from both x and x' , thereby enabling a comparison of concept distributions from which blindspots may be identified.

178 To define conceptual blindspots within the
 179 model g_θ , we must appraise the probability mass
 180 that the data-generating process apportions to a
 181 concept and juxtapose it with the mass appor-
 182 tioned by the model. To this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$
 183 and, availing ourselves of the accompanying text descriptions, constitute a set of synthetically gen-
 184 erated images $\mathcal{D}'_\mathcal{X}$. To estimate a concept’s probability of occurrence, we posit an **Energy model**
 185 $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional vector, each dimension of which affixes a
 186 scalar denoting the energy in the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. Such estimates are rendered
 187 tractable by our assumption that the data-generating process is invertible. Correspondingly, the
 188 probability that the data-generating process associates with the image $\mathbf{x}$ may then be defined as
 189 $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (where Z_ξ is the partition function), whence follows a population-level
 190 estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Building upon this quantity and the sigmoid map $\sigma(\cdot)$, we define
 191 below the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$.

192 **Definition 2 (Energy Difference).** Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-
 193 generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let
 194 $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the
 195 energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

196 where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
 197 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

198 Thus, the energy difference in the k^{th} concept characterizes the ratio between the probability that a
 199 concept occurs within a set of observations (here, $\mathcal{D}'_\mathcal{X}$) and that within a baseline dataset (here, $\mathcal{D}_\mathcal{X}$).
 200 On the strength of this measure, we may now define conceptual blindspots as follows.

201 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
 202 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
 203 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

204 In sum, we define a conceptual blindspot as a concept whose likelihood of occurrence in generated im-
 205 ages diverges markedly—whether through suppression or through exaggeration—from its prevalence
 206 under the data-generating process. Suppressed concepts evince disproportionately low activation (e.g.,
 207 $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout
 208 our analysis, we adopt the threshold values $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ so as to demarcate these
 209 regimes.

210 We further observe that this definition bears kinship with the notion of “mode collapse” examined in
 211 prior work (e.g., see Bau et al. [39]), the distinction residing in the granularity at which the analysis
 212 is conducted. Specifically, mode collapse concerns itself with exaggerated or suppressed odds of
 213 generating *entire images*, whereas our focus falls upon the altered odds of specific concepts. By way
 214 of illustration, should a model fail to produce images of an object set against a `white` background,
 215 we deem this concept a suppressed conceptual blindspot.

216 4 Method: Operationalizing the Definition of Conceptual Blindspots

217 We turn next to our pipeline for identifying conceptual blindspots within a generative model g_θ .
 218 In accordance with Sec. 3, the salient objects requisite to this end are twofold: (i) a set of images
 219 sampled from g_θ that permits comparison with the ground-truth generative process, and (ii) an energy
 220 model by which such comparison is effected. In what follows, we let $\|\cdot\|_F$ denote the Frobenius
 221 norm and $\|\cdot\|_0$ the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$,
 222 the relation $\mathbf{X} \geq 0$ connotes element-wise non-negativity. For $n > 0$, we set $[n] := \{1, \dots, n\}$ and
 223 denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

224 **From Prompts to Latent Representations.** To identify conceptual blindspots within a model g_θ ,
 225 we juxtapose a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs $(\mathbf{x}, \mathbf{t})$ sampled from the data-generating process G

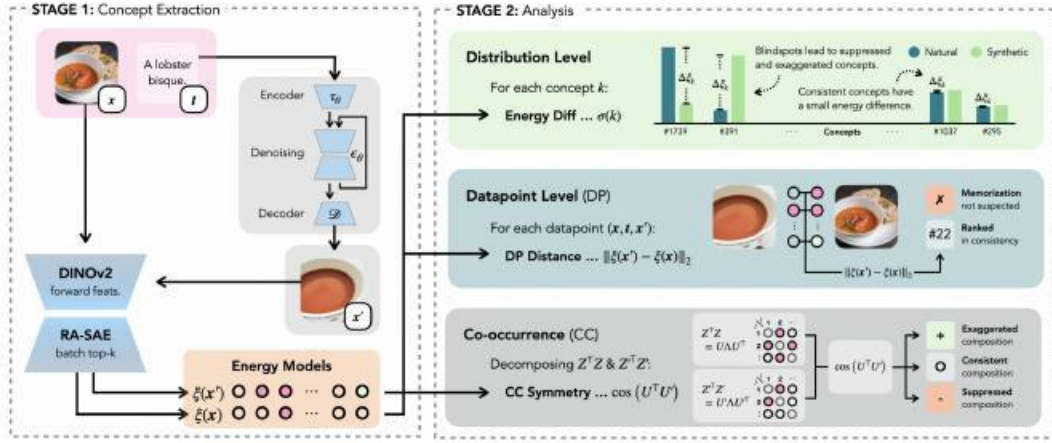


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , the concepts residing in x and t are extracted by first obtaining each image’s DINOv2 features, which are thereafter processed by a RA-SAE into sparse concept embeddings, thereby yielding the energy models $\xi(x)$ and $\xi(x')$ against their synthetic counterparts, drawn from the generative model g_θ by way of the accompanying text descriptions. In particular, $\xi_k(x) = E(c_k)$ encodes the energy in the k -th concept.

Specifically, given t , we synthesize a counterpart image x' by means of a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, instantiated as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling occurs in latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ conform to the standard cosine noise schedule. The terminal latent γ_0 is decoded through a pretrained VAE so as to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. Throughout the remainder of the paper, we treat g_θ as a black box mapping prompts to images: $t \mapsto x'$.

Defining the Energy Model. Building upon prior work that evidences the capacity of self-supervised learning methods to invert the data-generating process and recover the energy function underlying it up to linear transformations [126, 128, 133, 134], we adopt DINOv2 [135]—a state-of-the-art self-supervised model—as the substrate for our analysis. Under the supposition that the number of concepts underlying the DGP exceeds the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) upon its features so as to disentangle the subspaces corresponding to these concepts [50, 56, 137–139]. The underlying intuition is that, provided the concepts underlying the generative process are represented by DINOv2 along approximately orthogonal directions (as posited by our independence constraint in Def. 1), an SAE ought to be capable of isolating these concepts along individual dimensions of its latent space [136]. The activation that the SAE assigns to a given dimension will thereupon serve as our approximation to the ground-truth energy function ascribed to the concept modeled by that dimension.

Formally, letting $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote our feature extraction module (i.e., the DINOv2 model), we extract features $a = f(x) \in \mathbb{R}^d$ from both the natural and the synthetic images comprising datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$. Denoting by n the count of image-text pairs, we stack the real and generated features into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms by means of an SAE. Specifically, let $\mathbf{D} \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the activation matrix $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, wherein each row $z_i = \Psi(a_i)$ constitutes the concept decomposition of an image. The decoder reconstructs features via $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize the reconstruction error subject to constraints of sparsity and non-negativity:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \quad \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs are prone to drift toward arbitrarily oriented dictionaries, thereby rendering downstream analyses acutely sensitive to the random seed. To temper this instability and to render our study

reproducible and seed-invariant, we adopt the *Archetypal* SAE (RA-SAE) [56] in conjunction with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary $\mathbf{D}$ to be a convex combination of the training data. Concretely, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K',n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K',n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Consequently, every atom $\mathbf{D}_i$ lies within the convex hull of the data $\text{conv}(\mathbf{A})$, while any reconstruction $\mathbf{Z}\mathbf{D}^\top$ resides within the conic hull of the data $\text{cone}(\mathbf{A})$. This guarantees that the learned concepts remain faithful to the support of the data distribution [56]. Once trained, the SAE furnishes a consistent set of sparse codes: $\mathbf{Z}$ for real images and $\mathbf{Z}'$ for their generated counterparts. *These codes capture the selfsame prompt-conditioned visual semantics through shared, interpretable concepts, the activation value of each concept serving as the energy value that underpins our analysis of conceptual blindspots.* In summary, then, our method delineates a structured pipeline that, given a prompt and its associated real image (t, x) , produces two sparse concept vectors (z, z') , thereby enabling a direct comparison of the real and generated visual content within a common conceptual basis.

This recapitulates our full pipeline: commencing from a captioned image (t, x) , we synthesize a counterpart x' and map both images into a shared, sparse concept space by way of a vision encoder and a trained SAE, thus yielding interpretable representations (z, z') that will constitute the foundation for evaluating the conceptual shifts induced by the generative process.

5 Results

We analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—drawing upon $|\mathcal{X}| = 10,000$ image-text pairs together with their corresponding generations (Fig. 4). Our analysis unfolds across three levels (Fig. 3): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces failures attributable to ambiguity, omission, and memorization; and a **compositional** analysis exposes subtle distortions in the geometry of concept co-occurrence.

Our core contribution is an interactive exploratory tool, portrayed in Fig. 5. The tool is a web-based interface organized around a UMAP projection of concept representations, thereby affording the visualization and comparison of concept-level energy differences. It is made publicly available at a software repository, alongside pre-computed energy-difference data for the four models evaluated herein (SD 1.5, SD 2.1, PixArt, and Kandinsky). Every subsequent analysis in this paper proceeds from insights that this tool renders accessible. Its principal functionalities, which underpin these analyses, comprise:

- **Contrast different models and architectures.** For each evaluated model, the tool furnishes a UMAP visualization spanning the full complement of 32,000 concepts derived from the RA-SAE. Each scatter point denotes an individual concept, color-coded according to its energy difference.
- **Inspect concepts.** Every concept is endowed with a card presenting key statistics, representative real and generated images (x, x') , and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond the UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, thereby foregrounding the most salient conceptual blindspots.

Rather than exhaustively scrutinizing any single phenomenon, we present high-level findings that attest to the tool’s versatility and that invite broader, customizable exploration.



Figure 4: **Representative image-caption pairs** drawn from LAION-5B, accompanied by images generated from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

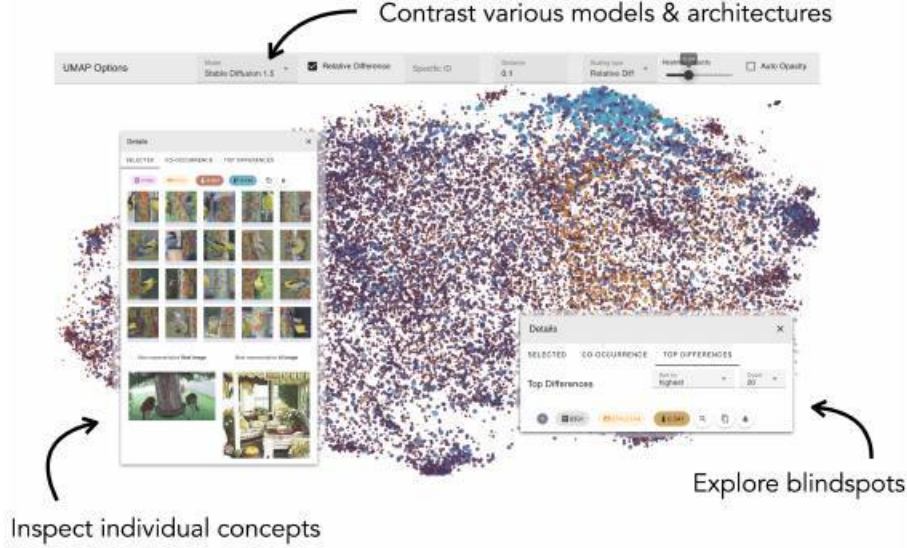


Figure 5: **Overview of the Exploratory Tool.** The web interface displays a UMAP projection for each evaluated model, wherein each dot denotes a concept, color-coded by its energy difference. Upon selection of a concept, a detail panel presents illustrative images, statistics, and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences is displayed alongside global rankings of blindspots.

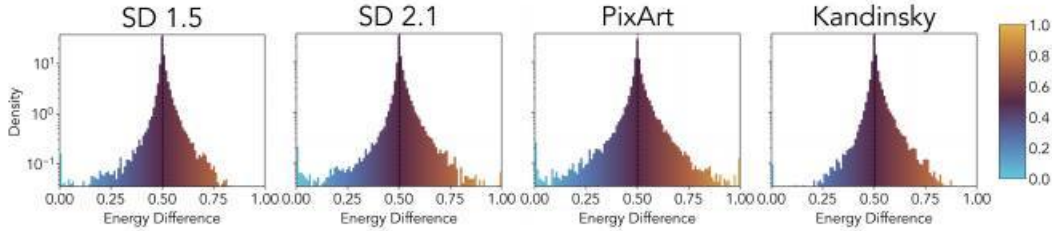


Figure 6: **Concept Energy Distribution.** Log-scale histograms of the energy differences $\delta(k)$ across 32,000 concepts, juxtaposing the natural and synthesized distributions for each evaluated model. Values to the left of zero denote suppressed concepts (under-represented), whereas values to the right denote exaggerated concepts (over-represented).

5.1 Generative Image Models Cannot Produce Unseen Concepts

To gauge the disparities between the generative models and the natural image distribution, we begin by analyzing the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by RA-SAE. As defined in Sec. 4, this quantity encapsulates the relative prevalence of each concept in the synthesized as against the natural image sets. A value of $\delta(k) < 0.1$ signals that concept k is under-represented (suppressed) among the generated images, whereas $\delta(k) > 0.9$ signals over-representation (exaggeration). Fig. 6 presents the distribution of $\delta(k)$ for each of the four evaluated models. Across all models, we discern heavy-tailed histograms bearing substantial mass at both extremes, a pattern indicative of systematic discrepancies in concept coverage. Tellingly, the left tail—corresponding to suppressed concepts—is at once denser and more elongated than its right-hand counterpart, betokening a consistent propensity toward concept suppression. This asymmetry finds expression in the negative skewness of the distributions: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

We further remark that, although all models manifest both suppressed and exaggerated concepts, their particular profiles diverge. PixArt, for instance, exhibits a wider spread, symptomatic of a more heavily suppressed concept distribution. The consistent left-skew across all distributions nonetheless underscores a shared inclination toward concept omission, even as the precise characteristics of this behavior demand further analysis—an inquiry we take up in the ensuing Sections.

326 5.2 Structure and Specificity of Conceptual Blindspots

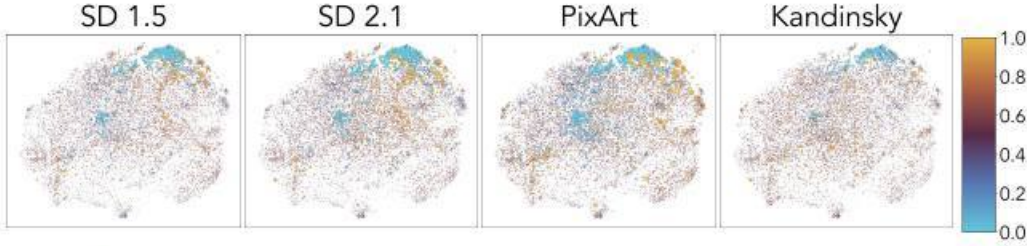


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored in accordance with their energy difference $\delta(k)$ between the natural and synthesized distributions. The emergent clusters disclose patterns of conceptual blindspots, with suppressed concepts occupying the blue extremity and exaggerated ones the yellow.

Whereas the preceding section quantified marginal discrepancies in concept frequency, here we interrogate their global structure by embedding the full set of 32,000 concepts into two dimensions—applying UMAP to the sparse codes and coloring the concepts by their $\delta(\cdot)$ values. As Fig. 7 attests, distinct clusters of concepts emerge across all models. These clusters frequently coincide with contiguous regions of conceptual blindspots, most conspicuously for suppressed (blue) concepts, intimating that blindspots are markedly structured—a reflection of biases shared across training distributions or architectural priors. To ascertain the consistency of these blindspot patterns across models, we further scrutinize both the magnitude and the structure of the concept-level $\delta(\cdot)$ values. Fig. 8 presents scatter plots together with pairwise Pearson correlation coefficients between the $\delta(k)$ vectors of SD 1.5 and every other model. As anticipated, SD 1.5 and 2.1 evince a strong correlation ($r = 0.82$), a corollary of their shared architectural and training pipelines. By contrast, their correlations with PixArt and Kandinsky prove substantially weaker— $r = 0.41$ and $r = 0.46$, respectively—signaling that these models accentuate distinct regions of the conceptual space.

In sum, the foregoing analysis reveals that whereas certain blindspots are universally shared—presumably by virtue of properties inherent to the dataset—others are markedly model-specific, arising from idiosyncrasies in training dynamics or model capacity. This dichotomy motivates the imperative, pursued in subsequent sections, to identify and study both the blindspots shared across models and those unique to particular models.

345 5.3 Qualitative Examples of Blindspots

We next visualize specific instances of both suppressed and exaggerated blindspots so as to discern which concepts fall within these regimes. In Fig. 9a, we present an exemplar of a conceptual blindspot suppressed by all models: we find that every evaluated model fails to reproduce the concept solid white on documents. As the figures make manifest, notwithstanding the caption’s explicit reference to this concept, none of the generated images register the intended visual semantics—a pattern suggesting that this region of the concept space is systematically under-sampled across models. Meanwhile, Fig. 9b foregrounds a model-specific blindspot: the concept pan is faithfully captured

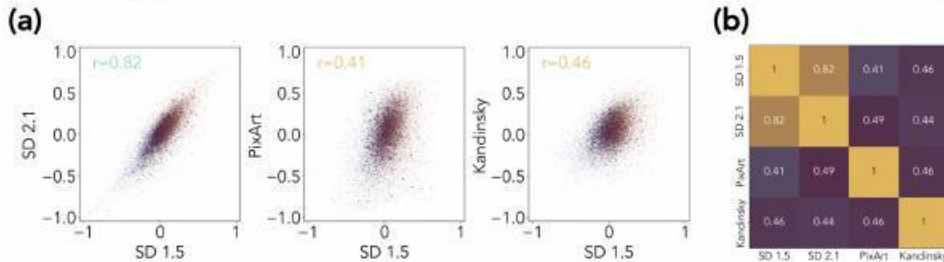


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with the Pearson correlation coefficients reported at top left. The strong alignment between SD 1.5 and SD 2.1 stands in contrast to the weaker correlations among the remaining architectures, a disparity indicative of model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** Heatmap of the pairwise Pearson correlation coefficients for δ between all models, quantifying the extent to which conceptual blindspots are shared across them.

by three of the models yet conspicuously absent from the generations produced by Kandinsky. This observation corroborates the findings of Sec. 5.2 wherein cross-model agreement was found to be pronounced in some cases yet circumscribed in others.

Conversely, in Fig. 10 we present a case of exaggeration, wherein the concept shadow under animal is unduly accentuated in the generated images. While shadows are in themselves mildly plausible, their consistent and pronounced rendering across models—set against the more nuanced and variable occurrences observed in natural images—bespeaks an overactive prior. Intriguingly, notwithstanding our efforts to locate concepts uniquely exaggerated by a particular model, we identified no clear instances thereof—a finding that suggests exaggerations are approximately universal.

Taken together, the foregoing examples concretely demonstrate how conceptual blindspots manifest in generated outputs, thereby illustrating that our energy-based diagnostic is capable of surfacing both shared and model-specific failure modes. Of particular note, it enables the identification of surprising model limitations—such as the consistent failure, across all models, to reproduce clear or solid background elements like whitespaces on documents. This raises the prospect that certain failure patterns may originate in architectural constraints or training-data biases that transcend the idiosyncrasies of any individual model.

Informative as these aggregate-level analyses are, they invite a more searching question: do these blindspots arise solely in the aggregate, across many samples, or do they manifest even at the level of individual datapoints? This finer-grained vantage permits us to probe the mechanisms underlying blindspots more directly—bringing to light cases of prompt misinterpretation, latent memorization, or both.



Figure 10: **Example of an Exaggerated Conceptual Blindspot.** Four synthesized images $\mathcal{D}'_{\chi}$, archetypal of the concept shadow under animal, alongside the corresponding natural image $\mathcal{D}_{\chi}$ and the caption t from which they were generated. Note the accentuated, nearly solid black shadows beneath the animals. We deem this blindspot universal inasmuch as it is present across all evaluated models.

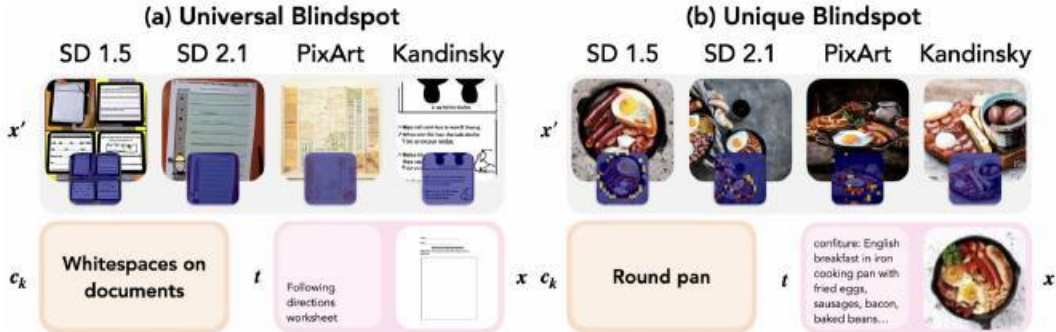


Figure 9: **Examples of Suppressed Conceptual Blindspots.** The natural images x , together with representative c_k and t , are displayed alongside four synthesized images x' generated by S_{θ} . The universal blindspot is present across all evaluated models, whereas the unique blindspot manifests solely in Kandinsky.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To advance beyond population-level statistics, we examine individual natural-versus-generated image pairs for which the $\delta(\cdot)$ values, averaged across all concepts, exhibit the largest and the smallest differences. This analysis facilitates a ready understanding of model successes and failures, the latter of which, we find, frequently issue from prompt ambiguity or from memorization artifacts. By way of example, Fig. 11a displays instances exhibiting near-zero difference in the average $\delta(\cdot)$ values. In such cases, the generated images are conceptually indistinguishable from the originals. Qualitative inspection nonetheless makes plain that this arises not because the model faithfully apprehends the prompt semantics, but rather through the outright replication of memorized templates: we observe repetitive visual structures (e.g., outlines of clothing or object arrangements), an indication that the model may be reproducing overly frequent patterns present in the training data. By contrast, Fig. 11b illustrates samples numbering among the largest $\delta(\cdot)$ values. These bespeak pronounced conceptual divergence between the synthesized and the natural image. While some of these discrepancies are attributable to underspecified or noisy captions, others lay bare genuine blindspots: the prompt describes a clear concept faithfully present in $\mathcal{D}_X$, and yet the model fails to realize it in $\mathcal{D}'_X$. This failure intimates that, even where language grounding is adequate, certain concepts lie beyond the model’s generative reach.



Figure 11: Datapoint-level Conceptual Alignment. (a) Examples exhibiting minimal energy differences, wherein the models appear to memorize training patterns. (b) Examples exhibiting large differences, wherein pronounced concept divergences arise from prompt ambiguity or from model limitations.

5.5 Analyzing Post-Training Effects

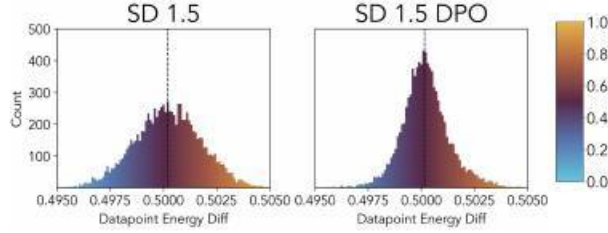


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of the datapoint-wise energy differences between the synthesized and the natural distribution for SD 1.5 models with and without DPO.

Post-training protocols—safety fine-tuning among them—have been argued to attenuate the diversity of model generations [140]. Given our pipeline’s capacity to isolate salient differences between a model’s generations and the ground-truth DGP, we next enlist it to elucidate the effects of DPO—a widely adopted safety fine-tuning protocol [141]. Specifically, we contrast two variants of the SD 1.5 model: one trained with DPO and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits at once a lower median and a narrower spread, betokening a distribution of generated concepts more consonant with the ground-truth DGP. This intimates that DPO may operate to regularize the model’s concept distribution, thereby encouraging outputs that more closely mirror the semantic content of the observed inputs. Although our analysis does not disentangle the specific inductive biases that DPO introduces, these results furnish empirical evidence that its optimization objective—which privileges human-preferred generations—indirectly fosters a closer accord with the training distribution. In particular, it curtails *both the over- and the under-*activation of individual concepts relative to the natural baseline. These findings underscore the utility of our pipeline in characterizing the downstream ramifications of post-training interventions—not merely with respect to output quality, but with respect to how such interventions reconfigure the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

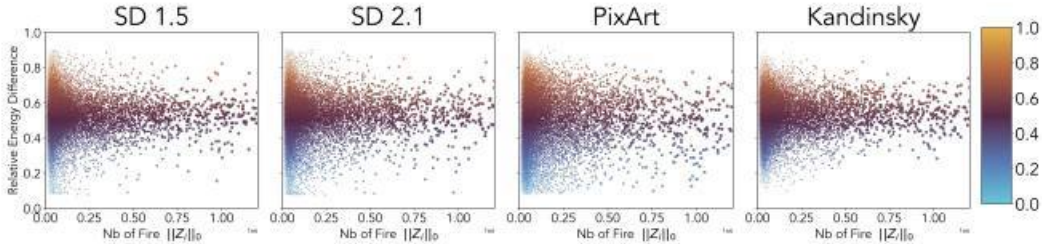


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots depicting the relationship between concept frequency (x-axis) and energy difference (y-axis) across the four evaluated models. Each point denotes a concept, its size being proportional to its activation frequency.

We hypothesized earlier that conceptual blindspots are not merely architectural artifacts but may also emerge as a direct consequence of the distributional peculiarities of certain features. In this section, we subject this hypothesis to empirical scrutiny by examining whether concepts that are seldom activated in natural images—that is, those inhabiting the long tail of the data distribution—give rise to blindspots in generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, where $\mathbf{Z}_{:,i}$ denotes the activations of concept i across the entirety of our images. We thereupon correlate this quantity with the absolute energy difference observed across the generated outputs. Fig. 13 visualizes this relationship for all evaluated models. We find that concepts of higher frequency in natural data tend to display lower energy discrepancies, whereas rare concepts—suppressed ones in particular ($\delta(k) < 0.5$)—exhibit substantial alignment errors. This pattern suggests that many blindspots stem

448 not from randomness or model idiosyncrasy, but from systematic effects bound up with long-tail
449 concept distributions. Redressing these deficiencies may demand not architectural modifications
450 alone but also strategies such as data reweighting or augmentation.

451 6 Discussion

452 Our analysis brings to light a multiplicity of conceptual blindspots across four widely adopted
453 generative image models. The results reported herein, however, do no more than skim the surface:
454 each individual finding could warrant an investigation of its own. Rather than probing any one of these
455 questions in depth, we instead showcase the versatility of our method and exploratory tool. Out of the
456 box, they underwrite the systematic identification of concepts that models struggle to generate, the
457 detection of memorization artifacts, the discovery of datapoints furnished with inadequate captions,
458 the quantification of post-training effects, and the characterization of conceptual shifts across model
459 architectures. We thereby open space for follow-up work to extend the depth of the analysis, the
460 breadth of evaluated architectures, and the inquiry into the root causes of conceptual blindspots.
461 Future work might likewise explore hierarchical representations of concepts within RA-SAE, so as to
462 admit a more nuanced analysis.

463 **Limitations.** We wish to bring several limitations of our work into relief. In predicated concept
464 extraction and representation upon DINOv2 and RA-SAE, our approach is inherently circumscribed
465 by the kinds of concepts these models are able to capture; concepts they represent poorly will elude
466 our analysis. Furthermore, substantial though our sample of 10,000 images is, it may not fully
467 encompass the long tail of rare concepts, concept co-occurrence, or other compositional statistics.

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Appendix

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A Experimental Setup

This section sets out in detail the experimental setup underpinning our analysis of four widely adopted generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or upon its subsets and derivatives. The code is made available at a software repository.

A.1 Observation Space

The observation space is constituted by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which furnishes our domain of natural images. Owing to concerns surrounding CSAM and other unsafe content within the dataset, the original data release is no longer accessible. We accordingly employ a substitute release comprising a subset of this dataset subjected to additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three steps: (1) loading the full LAION dataset by means of the Hugging Face datasets library; (2) performing validation, via HTTP HEAD requests, to confirm proper URL structure and resource availability; and (3) drawing random samples with replacement until the target count of 10,000 valid samples is attained. This approach, which yields D_G populated with (x, t) tuples, ensures that our observation space comprises accessible image-text pairs amenable to comparative analysis—notwithstanding that a substantial fraction of the underlying image URLs has been rendered unavailable since the original release. Additional examples of synthesized images are presented in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} standing in one-to-one correspondence with D_G , thereby yielding triplets (x, x', t) . Specifically, given the 10,000 image-text pairs (x, t) drawn from D_G , we employ t to synthesize counterpart images x' by means of each generative model g_θ .

The synthesis process adheres to the standard text-to-image generation pipeline appropriate to each model architecture, implemented via the Hugging Face diffusers library, with the models loaded at mixed precision (fp16). All synthetic images are generated at a resolution of 512×512 pixels under default parameters.

Stable Diffusion 1.5. We employ the checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (itself a mirror of the deprecated <https://huggingface.co/ruwnayml/stable-diffusion-v1-5>). Inference is conducted over 50 inference steps, with the guidance scale held fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (deployed in the analysis of Sec. 5.5) conforms to the baseline SD 1.5 implementation, save that it substitutes the UNet component with a DPO-trained version drawn from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We employ the checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference is conducted over 50 inference steps, with the guidance scale held fixed at 7.5.

Kandinsky. We employ the checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference is conducted over 100 inference steps, with the guidance scale held fixed at 4.0.

PixArt. We employ the checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference is conducted over 50 inference steps, with the guidance scale held fixed at 7.5.

940 A.3 • Distribution Level Analysis

941 **Section 5.1** We compute the energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated
942 model. During normalization, we apply the sigmoid transformation with temperature $T = 0.8$. The
943 resulting values are visualized as log-scale density histograms comprising 100 bins spanning $[0, 1]$.

944 **Section 5.2** We embed the complete set of 32,000 concepts into two-dimensional space by ap-
945 plying UMAP to the sparse concept codes. Each point within this UMAP denotes an individual
946 concept, colored in accordance with its energy difference $\delta(\cdot)$, so as to accentuate both suppressed
947 and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson
948 correlation coefficients between the $\delta(\cdot)$ vectors of all model pairs, generating thereby both scatter
949 plots and correlation matrices. This analysis discloses whether blindspots cluster within conceptual
950 space and distinguishes model-specific from universal patterns of conceptual blindspots.

951 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine
952 the extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select
953 concepts for which $\delta(\cdot) < 0.1$; for exaggerated blindspots, we select those for which $\delta(\cdot) < 0.9$. The
954 presented examples are manually annotated with textual descriptions of their respective concepts,
955 arrived at through inspection of the most activating images and the attendant spatial attention patterns.
956 We delineate our ongoing efforts to automate this concept interpretation in Appendix D

957 **Section 5.5** We compare SD 1.5 with and without DPO in the following manner: for each image
958 pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors,
959 $\|\xi(x') - \xi(x)\|_2$. Prior to taking their mean, we apply a sigmoid transformation with temperature
960 $T = 0.8$ to the element-wise differences. This procedure yields datapoint-wise energy differences
961 that quantify the extent to which each generated image departs from its natural counterpart within
962 concept space. Finally, these differences are visualized as overlapping histograms that contrast the
963 two model variants.

964 **Section 5.6** For each concept c_k , we count its empirical frequency $\|Z_{:,i}\|_0$ (the tally of non-zero
965 activations across the natural dataset). A sigmoid normalization with temperature $T = 0.4$ is thereafter
966 applied to the energy differences $\delta(\cdot)$. The analysis is visualized by way of scatter plots in which
967 the x-axis denotes the empirical concept frequency and the y-axis the sigmoid-transformed energy
968 difference. Point sizes are proportional to activation frequency, while point colors are proportional to
969 the magnitude of the energy differences.

970 A.4 • Datapoint Level Analysis

971 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
972 their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, thereby obtaining a scalar measure of conceptual
973 divergence for each image pair. The samples are then ranked by their energy differences. Minimal
974 divergence is indicative of potential memorization artifacts, whereas maximal divergence points to
975 pronounced conceptual failures. This analysis enables the qualitative inspection of specific failure
976 modes.

977 A.5 • Co-occurrence Analysis

978 **Appendix F** For both the natural and the synthesized data, D_G and $D_{g\theta}$, concept co-occurrence
979 patterns are analyzed by way of the co-activation matrix $Z^T Z$, which encodes the pairwise correla-
980 tions in concept usage. Spectral analysis, effected through eigendecomposition, is performed so as
981 to examine the dominant conceptual directions. The alignment between the natural and synthetic
982 co-occurrence structures is assessed by means of cosine similarity heatmaps computed between
983 the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are
984 visualized as square heatmaps, wherein perfect diagonal alignment would betoken identical principal
985 concept axes, while off-diagonal patterns would disclose rotations and mismatches in compositional
986 geometry.

B Custom RA-SAE Training Details

This section recapitulates the training details of our custom archetypal SAE (RA-SAE) [56]. The model is open-sourced at a software repository.

Dataset. The auto-encoder is trained upon the complete ImageNet-1k training split (≈ 1.28 M RGB images). Each image is converted into 261 visual tokens by means of DINOv2 [135], and the tokens are furnished to the SAE bereft of class or position embeddings. The aggregate number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary is endowed with 32,000 concept dimensions. For the sparse activation rule, we adopt top- k masking with $k = 5$; activations lying outside the largest five per input are set to 0. The weights are initialized by way of the Xavier/Glorot scheme. Training is conducted at mixed precision (fp16), the final ten epochs being performed at full precision.

Optimizer and Schedule. The model is trained for 50 epochs by means of the base AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$), with weight decay fixed at 10^{-5} . Linear warm-up is applied over the first 5% of steps, succeeded by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. An MSE loss is employed in concert with an auxiliary term that penalizes activations never entering the top- k set, with $\lambda = 10^{-5}$.

1003 C Computational Resources

1004 This section summarizes the GPU resources marshalled for the training and experiments in support of
1005 this paper. In aggregate, we expended approximately 202 GPU-hours on NVIDIA H100s and H200s.

1006 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

1007 **Synthesized Images.** Generating the full D_{g_θ} (see Appendix A) required roughly 5 hours per
1008 generator when distributed across four NVIDIA H200 GPUs. Across five generators, this amounted
1009 to approximately 100 GPU-hours on an NVIDIA H200.

1010 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level consumed
1011 about 3 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

1012 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
1013 datapoint, and ranking datapoints per concept likewise required approximately 3 hours per generator
1014 on a single H200 GPU (total ~ 15 GPU-hours).

1015 D Autointerpretability

1016 In an endeavor to automate the interpretation of the concepts identified by the RA-SAE, we qualita-
 1017 tively appraised the capacity of Vision-Language Models (VLMs) to describe them. Specifically, we
 1018 queried ChatGPT 4o via the OpenAI API using a variety of prompts and formats of an image x'' in
 1019 which the given concept was salient, recording in each case the description that the VLM generated.
 1020 In this section, we bring the findings of our exploration into relief. For the purposes of evaluation, we
 1021 permitted the VLM to furnish its full reasoning; to render this a scalable solution, a straightforward
 1022 modification of the prompt—instructing the VLM to supply the description alone—would suffice to
 1023 filter out the reasoning.

1024 First, we provided the VLM with two images: a raw rendition of x'' and a version of x'' upon which
 1025 a heatmap was superimposed so as to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

1026 As the example makes evident, the VLM did not construe the heatmap visualization correctly. In lieu
 1027 of describing the wood texture in the background—the genuinely salient region—it described the
 1028 ring in the foreground, which received no attention whatever from the RA-SAE.

1029 To ascertain whether this constituted a prompting deficiency, we revisited the same scenario—this
 1030 time availing ourselves of an expanded prompt that explicated the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

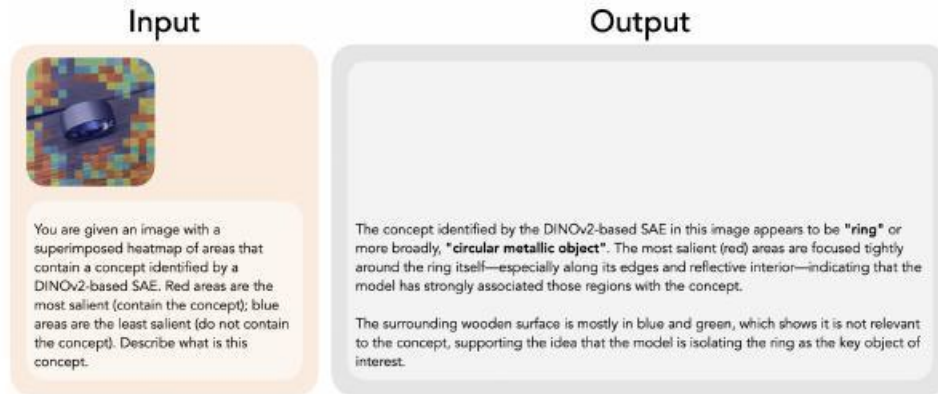
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

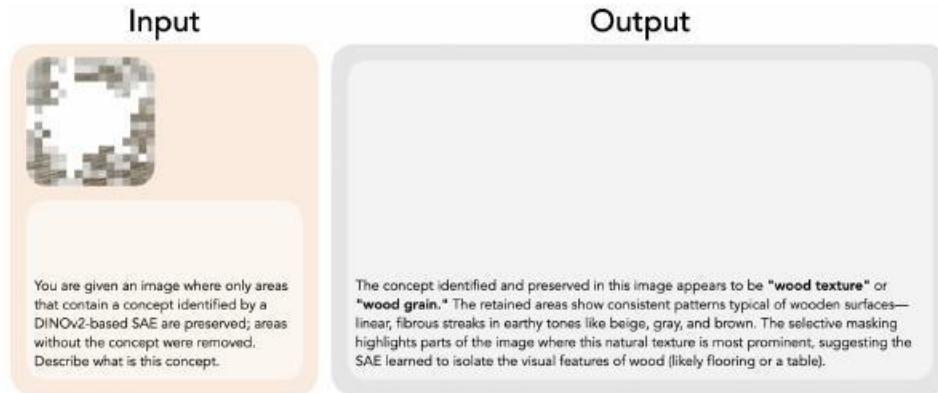
1031 This modification of the prompt did not resolve the issue, for the VLM persisted in identifying the
 1032 ring as the prominent concept within the image. Several further prompting techniques—among
 1033 them those describing the heatmap in still greater depth (e.g., by including the code used for its
 1034 generation)—were explored, with comparable results.

1035 We next examined the effect of providing the VLM solely with the heatmap-superimposed x'' ,
 1036 accompanied by a similar prompt explaining how the heatmap ought to be interpreted:



1037 Once again, the result failed to identify the wooden background as the prominent concept within the
 1038 image, fixating instead upon the ring. As in the preceding case, we experimented with a variety of
 1039 prompting strategies under this image format, yet the results remained largely unaltered.

1040 From these observations, we concluded that the heatmap in isolation does not suffice to direct the
 1041 VLM's attention toward the salient regions. The presence of less relevant regions appears to distract
 1042 the model. To remedy this, we substituted for the heatmap-superimposed x'' an alpha-masked version
 1043 of x'' :



1044 As the example demonstrates, the VLM was now able to identify wood texture correctly as the
 1045 dominant concept. We observed consistent success across a range of concepts and images.

1046 The formatting of x'' and the prompt exhibited above yielded the most reliable results in our qualitative
 1047 evaluation. We note, however, that this evaluation is circumscribed both by its qualitative character
 1048 (owing to the absence of ground-truth annotations) and by its concentration upon a single VLM. We
 1049 trust that future work on the autointerpretability of SAE concepts may build upon and extend this
 1050 analysis.

E Additional Results: Model-Specific Blindspots

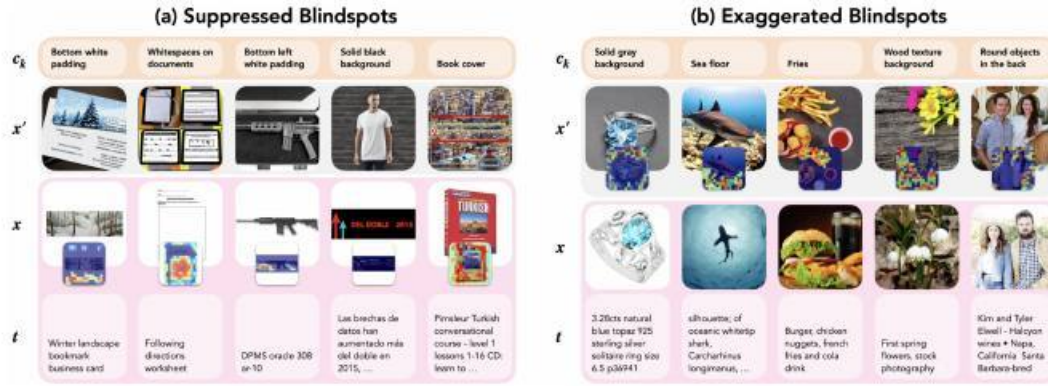


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, we display the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots), selected on the basis of the highest absolute activation. The spatial heatmap for the concept is superimposed atop the image.

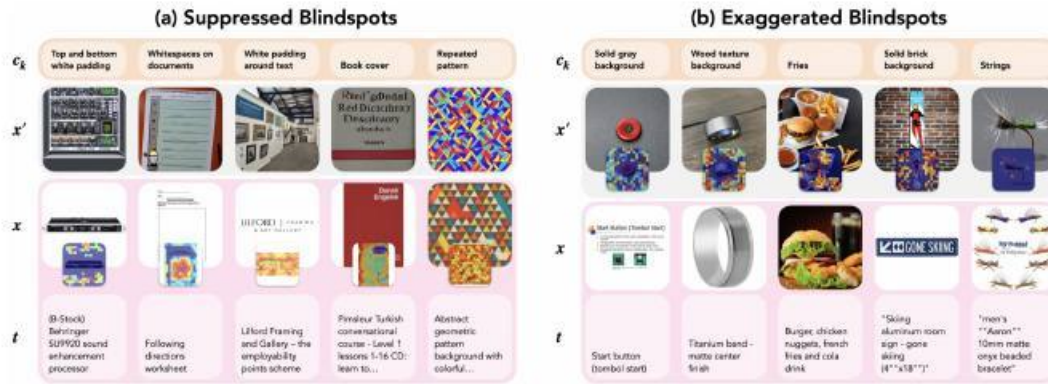


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, we display the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots), selected on the basis of the highest absolute activation. The spatial heatmap for the concept is superimposed atop the image.

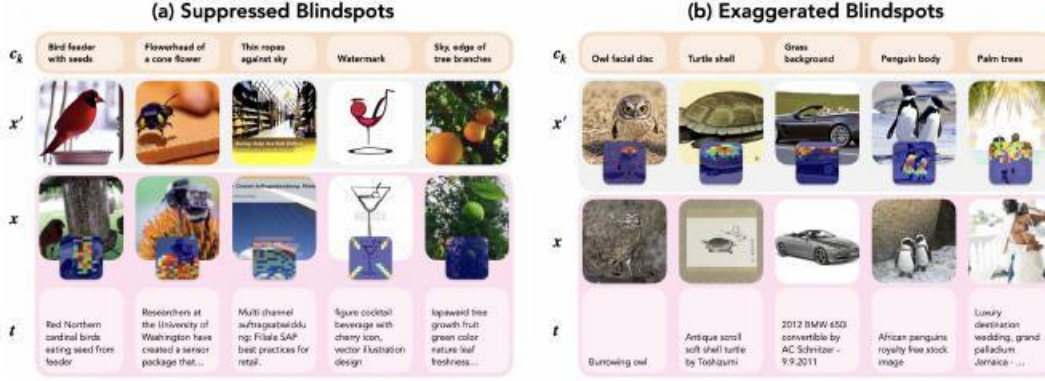


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, we display the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots), selected on the basis of the highest absolute activation. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, we display the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots), selected on the basis of the highest absolute activation. The spatial heatmap for the concept is superimposed atop the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has been centered upon individual concept activations. Yet visual scenes are seldom composed of isolated concepts; rather, they are organized through rich, structured co-occurrence patterns that encode compositional semantics. We now interrogate whether generative models apprehend this higher-order structure, doing so by analyzing the co-activation matrix $Z^T Z$, which registers the pairwise correlations in concept usage.

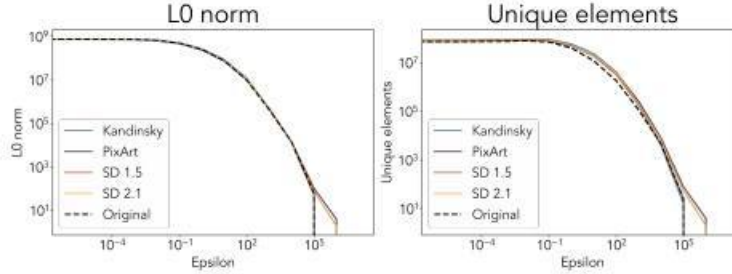


Figure 18: **Sparsity and Structural Divergence.** On the left: the L_0 norm of the co-occurrence matrix ZZ^T as a function of ϵ (threshold), indicating how many entries remain active within each model. On the right: the number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve the global sparsity structure yet diverge in their activation content.

Surprisingly, when appraised at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded at varying values of ϵ —tracks closely between the natural and synthesized distributions across all models. This attests that the gross connectivity of the conceptual graph—that is, which concepts tend to co-activate at all—is well preserved. Formally, one may regard $Z^T Z$ as the adjacency matrix of a weighted, undirected graph over concepts, wherein the edge weights encode co-activation strength across the dataset.

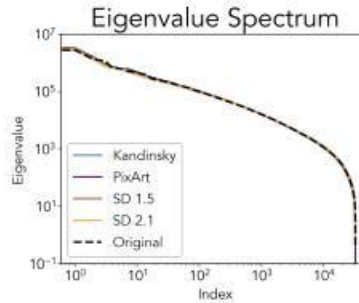


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T across the models and the natural distribution. All evaluated models reproduce the heavy-tailed decay characteristic of the natural distribution.

As illustrated in Fig. 18 (right), however, the specific content of these co-activations diverges: a substantial proportion of the entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that, while the capacity for compositionality is retained, the identity of the active pairings may shift—potentially in reflection of model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 displays the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. All spectra exhibit a heavy-tailed decay consonant with power-law behavior, indicating that generative models preserve both the overall rank structure and the allocation of variance across conceptual dimensions.

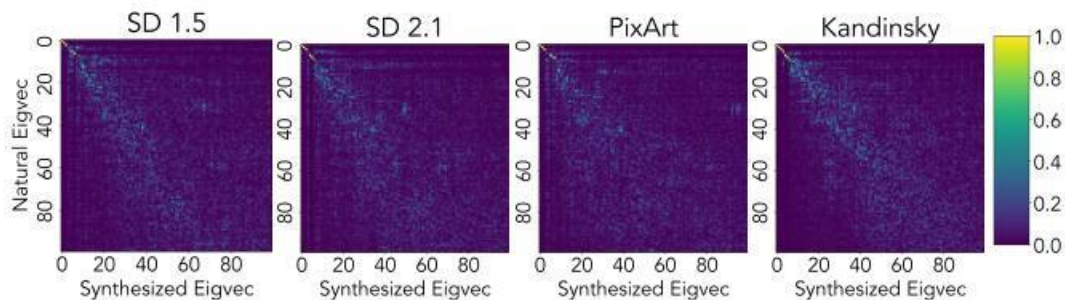


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . The diagonal structure evinces alignment of the dominant conceptual directions, the degree of which varies across the four models.

1076 Further, we examine the alignment of the dominant conceptual directions by way of cosine similarity
 1077 heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices
 1078 (Fig. 20). Although all models exhibit partial diagonal alignment—implying an overlap in the
 1079 principal concept axes—the off-diagonal entries lay bare rotations and mismatches in the higher
 1080 modes, betokening manifest deviations in compositional geometry.

1081 Taken together, these findings reveal that diffusion models approximate the global shape of concept
 1082 co-activation surprisingly well, and yet deviate in subtle and structured ways when scrutinized
 1083 through the spectral lens. Such higher-order discrepancies may well underlie the failures observed in
 1084 generating coherent, multi-object scenes or relational concepts.

1085 G Additional Results: Blindspot Stress Testing

1086 To stress-test the blindspots identified by our method, we assembled a range of prompts describing
 1087 these blindspots and enlisted them to generate a multitude of images. We thereupon contrasted the
 1088 outputs of models in which the concept had been identified as a blindspot against those in which it
 1089 had not.

1090 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1091 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1092 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1093 prompt and analyzed by means of our custom RA-SAE model (see Appendix B), which ranked them
 1094 according to the intensity with which the desired concept appeared. All images were subsequently
 1095 reviewed by hand to determine whether the blindspot had been successfully depicted.

1096 As the following examples of suppressed concepts attest, although certain aspects of the target concept
 1097 did occasionally surface (e.g., a holder or string for the bird feeder blindspot, and a round hole for
 1098 the glossy DVD disc blindspot), the models by and large failed to render the full concept. This
 1099 accords with our method’s assessment and lends support to the validity of the stress test.

1100 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated from various prompts involving the bird feeder concept in the course of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot—a designation borne out by the observed behavior: the model proves unable to generate a corresponding image. By contrast, SD 1.5, for which this concept was not identified as a blindspot, is able to generate it.



Figure 22: Examples of images generated from various prompts involving the glossy DVD disc concept in the course of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot—a designation borne out by the observed behavior: the model proves unable to generate a corresponding image. By contrast, Kandinsky, for which this concept was not identified as a blindspot, is able to generate it.

1102 H Additional Examples: Synthesized Images



Figure 23: Additional examples of image-caption pairs drawn from LAION-5B, accompanied by images generated from the identical prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1103 I Concentration bounds for δ

1104 In our experiments, we estimate $\delta(k)$ from $n = 10,000$ paired samples for each concept. Modest
 1105 though this budget is, it raises the natural question of whether it suffices to yield reliable estimates.
 1106 To adjudicate this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ by recourse
 1107 to McDiarmid's inequality [142]. The resulting bound is tight and establishes that, even with
 1108 comparatively few samples, we may obtain rapid and accurate estimates of concept bias.

1109 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1110 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1111 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1112 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1113 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1114 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1115 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1116 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1117 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1118 which proves the claim. □

1119 In practice, most concept scores $\delta(k)$ are sparse, the majority being concentrated near zero and only a
 1120 handful attaining values as large as 10. The concentration bound establishes that, even for the largest
 1121 observed biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that, with high probability,
 1122 deviate from the true value by no more than a small ε . This vindicates our sampling strategy and
 1123 confirms that accurate measurements of bias are attainable with limited data.

1124 J Monotonicity and Calibration-Free Interpretation of δ

1125 Our aim in analyzing blind spots is to rank concepts by the severity of their generative bias, a purpose
 1126 we serve in practice by means of the score $\delta(k)$. One may nonetheless wonder whether such a score
 1127 introduces distortions relative to more direct quantities, such as the energy gap or the odds ratio. The
 1128 following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and hence
 1129 inherits their ordering—thereby guaranteeing that no calibration is required when $\delta(k)$ is employed
 1130 to rank concepts.

1131 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1132 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1133 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1134 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1135 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1136 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1137 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1138 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1139 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1140 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1141 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1142 concepts.

1143

□

1144 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by the energy gap or by the
 1145 conceptual generation odds ρ_k . No calibration is necessary, and all three quantities preserve one and
 1146 the same total ordering over concepts.

1147 K Stability of FID Under SAE Embeddings

1148 In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID)
 1149 computed within the original activation space (of dimension d) and the FID obtained after the
 1150 application of a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we
 1151 assume that D possesses orthonormal columns while not being necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1152 We begin by recalling a simple fact: should D be non-overcomplete, orthogonal, and satisfy $k = d$,
 1153 then we have an isometry between A and Z , whence the FID is perfectly preserved. This case being
 1154 unrealistic, however, we shall subsequently turn to the overcomplete setting and demonstrate that
 1155 the FID may be bounded by the extreme singular values of D . We will work throughout with the
 1156 Wasserstein-2 metric $\mathcal{W}_2$, observing that the FID is merely $\mathcal{W}_2^2$ specialized to Gaussians.

1157 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
 1158 $\mu(D^{-1}z)$. We denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of
 1159 D —equivalently, the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1160 Empirically, one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof in no way relies upon this. We
 1161 commence with a simple lemma addressing the case in which D is not overcomplete.

1162 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1163 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1164 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1165 *Proof.* Orthogonality of D entails preservation of the Euclidean norm, and the push-forward com-
 1166 mutes with the map inside the $\mathcal{W}_2$ infimum; the integrand being unchanged, the value of the infimum
 1167 is identical. $\square$

1168 This case, however, is decidedly unrealistic, inasmuch as SAEs typically depend upon overcomplete-
 1169 ness in order to extract meaningful and interpretable concepts. In the overcomplete case, D is no
 1170 longer orthonormal, yet we may still avail ourselves of a column-orthonormal dictionary. We shall
 1171 exploit this property to demonstrate that a bound may be obtained by way of the extremal singular
 1172 values of $D^\top D$.

1173 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1174 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1175 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1176 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1177 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1178 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1179 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1180 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1181 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1182 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1183 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1184 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1185 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1186 finishes the proof. $\square$

1187 In essence, Theorem 6 appraises us that the application of a column-orthogonal overcomplete SAE
 1188 dictionary cannot distort the Fréchet Inception Distance by more than the square of its extremal
 1189 singular values. When D is nearly orthogonal – empirically, we typically found that $\sigma_{\min}, \sigma_{\max} \approx 1$
 1190 – the result implies that the FID measured within the SAE feature space is essentially proximate to the
 1191 canonical FID. $\square$

1192